Proposed Plan of Study

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Whose Contribution Counts? AI Mediation and Collective Authorship in Digital Improvisation

In collaborative creative tools, a generative model reinterprets what participants contribute, and its output may become more visible than the contribution that prompted it. My doctoral research asks how participants, facilitators and a generative-AI interface shape whether a person's contribution is recognised, carries creative authority, and is credited in a collective work. This is a question of AI-mediated communication: I study AI not as a creative partner but as a medium that translates one person's offer to others, and ask whether the system's record of who supplied what matches how collaborators recognise one another.

Table 1. Closest related work, as reported in each source.

Study	AI's role	What this project adds
Xu et al. 2026 (CHI)	Shared ChatGPT in one group task	Improvisation; credit across sessions
Trajkova et al. 2024 (CHI)	Proposed AI dance partner, not yet tested	An implemented mediator in group work
Epstein et al. 2020 (*iScience*)	AI art described to online observers	Credit claimed by the people making the work
Caines 2026 (*AJSE*)	GenAI tools in course design	Recognition of individual contributions

My objects of study are small-group digital improvisation workshops with adult improvisers, and the shared audiovisual environment I will build. A language model proposes transformations of participants' phrases into light, form and sound, which contributors or groups accept, revise or refuse.

Under the research-creation option, the environment, improvised works and workshop-tested scores form the creative component. Methods are scoping interviews, co-design with practitioners, repeated workshops (provisionally three groups, four sessions each) comparing contributor-led, group-led and human-mediated arrangements, and video-based reflection interviews. The framework is Lucy Suchman's account of how agency is figured at the human–machine interface.

The project belongs to Technology in Practice (minor: Media & Culture) and to the program's commitment to "the application of critical theory and research to contemporary issues and practices in communication and culture." I hope to work with Dr. Rebecca Caines, whose research on improvisation as methodology, socially responsive creative technologies, and generative AI as a critical improvisation practice fits this approach.

Each stage of my training asked how designed environments shape how people perceive and relate to one another. In architecture at Tabriz Islamic Art University (Best Dissertation Award, 2016), I designed space as a medium of social life, where thresholds and paths decide who gathers and is seen. Landscape architecture at Istanbul Technical University extended this to shared public space; in its Virtual Landscape Lab I studied how people perceive mediated environments (eCAADe 2023). At Simon Fraser University's iViz Lab, my thesis research moved from space to interfaces: I built multiplayer systems that let remote teammates sense one another's heart-rate state without emotion labels (ACM ISS 2026). Participants accepted the display only if they controlled who saw it, a communication question about who controls how a system represents people. The PhD carries it into AI-mediated creative communities.